Draft version February 22, 2020
Typeset using LaTeX default style in AASTeX62

Ay 102: Shocks and SN

Yuhan Yao

ABSTRACT

This is my notes on Ay102, winter 2020. I mostly follow instructions given in Dopita & Sutherland (2003), Kwok (2007), Draine (2011), and Tielens (2005). Vink (2012) gives a relatively recent review of X-ray properties of SNR.

1. EVOLUTION OF A SN REMNANT IN A UNIFORM ISM

The evolution of Supernova Remnants (SNRs) is usually divided in four phases: (I) the ejecta dominated phase, in which the mass of the supernova ejecta, $M_{\rm ej}$, dominates over the swept-up mas $M_{\rm swept}/$; (II) the Sedov-Taylor phase, for which $M_{\rm swept} > M_{\rm ej}$, but for which radiative losses are not energetically important; (III) the pressure-driven, or "snow-plow" phase, in which radiative cooling has become energetically important; the evolution of the shock radius is now best described by using momentum conservation; (IV) the merging phase, in which the shock velocity and temperature behind the shock become comparable to, respectively, the turbulent velocity and temperature of the interstellar medium. For a list of SNR see https://en.wikipedia.org/wiki/List_of_supernova_remnants.

Although these discrete phases provide a useful framework to think of SNRs, it should be kept in mind that it is an oversimplification, and the phase of an individual SNR is not always that easily labeled. Now, let us consider the simplest case of a spherically symmetric explosion of a star in a uniform medium with hydrogen density n_0 and temperature T_0.

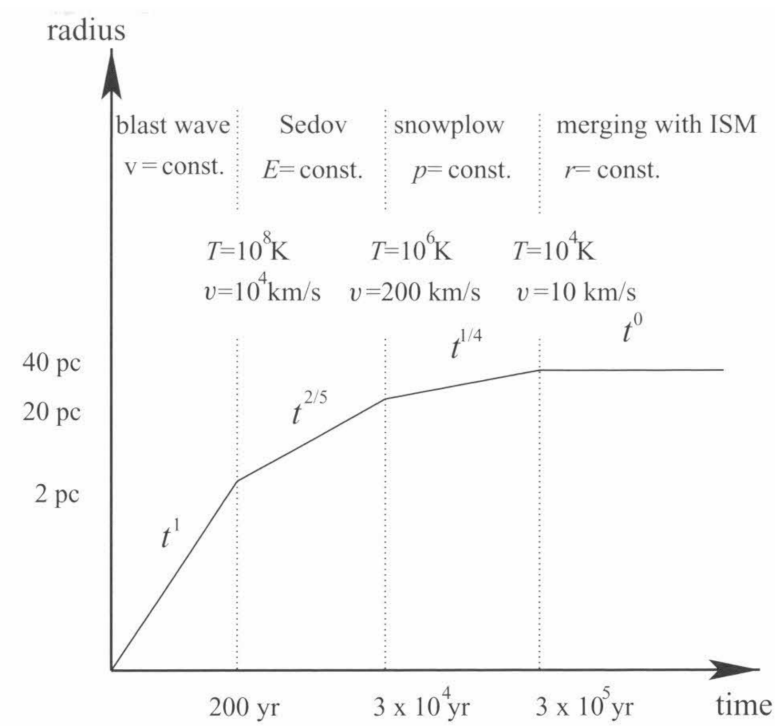

Figure 1. An illustration of the four different phases in the dynamics of a SNR. Fig 4.9 of Rosswog & Brüggen (2011).

1.1. *The Free-expansion Phase*

This phase is defined by the fact that the velocity of the ejecta is constant with time. Typical ejecta velocity is

$$v_{\rm ej} \approx \sqrt{\frac{2E_0}{M_{\rm ej}}} = 1.00 \times 10^4\,{\rm km\,s^{-1}} E_{51}^{1/2} \left(\frac{M_\odot}{M_{\rm ej}}\right)^{1/2} \tag{1}$$

This velocity is far greater than the sound speed in the surrounding material, and the expanding ejecta will therefore drive a fast shock into the circumstellar medium.

The expanding mass of ejecta creates a shock that travels outwards through the ISM (**blastwave**) and when the pressure of the shocked CSM exceeds the thermal pressure in the ejecta, a **reverse shock** is created. The outward traveling shock heats the ISM behind it to very high temperatures. Thermal bremsstrahlung is seen in X-rays, and synchrotron emission is seen in radio from ejected particles spiralling around μG magnetic fields. The reverse shock has a finite amount of material to propagate inwards, shock-heats the ejecta (which was previously been cooled by adiabatic expansion), and dies away.

The condition for being in this phase is that $M_{\rm ejecta} \gtrsim M_{\rm swept}$, that is, the mass swept up by the ejecta from the ISM is not a significant fraction of the ejecta mass. Therefore, the radius of the blastwave when this ends is

$$R_1 = \left(\frac{3M_{\rm ej}}{4\pi\rho_0}\right)^{1/3} = 5.88 \times 10^{18}\,{\rm cm} \left(\frac{M_{\rm ej}}{M_\odot}\right)^{1/3} n_0^{-1/3} \tag{2}$$

and the time when it ends is

$$t_1 \approx \frac{R_1}{v_{\rm ej}} = 186\,{\rm yr} \left(\frac{M_{\rm ej}}{M_\odot}\right)^{5/6} E_{51}^{-1/2} n_0^{-1/3} \tag{3}$$

As is shown in Figure 1, this phase lasts about 200 years.

1.2. *The Adiabatic Phase (also called Sedov-Taylor Phase)*

In this phase, the mass of the swept up ISM dominates the mass of the ejected material ($M_{\rm ejecta} < M_{\rm swept}$). However, the density of the material behind the shock is not yet large enough for cooling to be significant. Therefore, the gas expands adiabatically. The SNR now enters a phase that can be approximated by idealizing the problem as a "point explosion" injecting only energy E_0 into a uniform-density zero-temperature medium of density ρ_0.

The total energy of the expanding remnant is

$$E_0 = \frac{4\pi}{3} R^3 \rho_0 (E_{\rm int} + E_{\rm kin}) \tag{4}$$

where the kinetic energy per unit mass of the shocked gas is

$$E_{\rm kin} = \frac{1}{2} v_1^2 = \frac{1}{2}\left(\frac{2}{\gamma+1} v_s\right)^2 \tag{5}$$

Since $v_{\rm ej} \gg c_s$, we can make use of the strong shock approximation, $u_0 = -v_s$, and the internal energy per unit mass for the shocked gas is

$$E_{\rm int} = \frac{P_1}{(\gamma-1)\rho_1} = \frac{\left(\frac{2}{\gamma+1}\right)\rho_0 u_0^2}{(\gamma-1)\frac{\gamma+1}{\gamma-1}\rho_0} = \frac{1}{2}\left(\frac{2}{\gamma+1} v_s\right)^2 \tag{6}$$

Hence, writing v_s as $\dot{R}$,

$$E_0 = \frac{3\pi}{4} R^3 \dot{R}^2 \rho_0 \tag{7}$$

From dimensional analysis considerations, with the initial condition $R(0) = 0$, we can derive the Sedov-Taylor solution by integrating Eq (7) to get

$$\begin{aligned} R &= \left(\frac{25}{3\pi}\right)^{1/5} \left(\frac{E_0}{\rho_0}\right)^{1/5} t^{2/5} \\ &= 5.6 E_{51}^{1/5} n_0^{-1/5} \left(\frac{t}{10^3\,{\rm yr}}\right)^{2/5} {\rm pc} \end{aligned}$$

The expansion velocity can be found by taking the first derivative of $R(t)$:

$$\dot{R} = \frac{2}{5}\left(\frac{25}{3\pi}\right)^{1/5}\left(\frac{E_0}{\rho_0}\right)^{1/5} t^{-3/5}$$
$$= 2200 E_{51}^{1/5} n_0^{-1/5}\left(\frac{t}{10^3\,\mathrm{yr}}\right)^{-3/5}\ \mathrm{km\,s^{-1}}$$

In an ideal gas, each particle contributes equally to the pressure. Assuming that the remnant is totally ionized, we see that the internal energy of the shocked gas is

$$E_{\mathrm{int}} = \frac{3}{2}\frac{kT}{\mu m_p} \tag{8}$$

Let this equal to Eq. 6, we get

$$T = \frac{2(\gamma-1)}{(\gamma+1)^2}\frac{\mu m_p}{k}u_0^2 = \frac{3}{16}\frac{\mu m_p \dot{R}^2}{k} = 10^8 \mu E_{51}^{2/5} n_0^{-2/5}\left(\frac{t}{10^3\,\mathrm{yr}}\right)^{-6/5}\ \mathrm{K} \tag{9}$$

When $t = 10^4\,\mathrm{yr}$, $\mu = 0.6$, $E_{51} = 1$, $n_0 = 1$, $T = 3.8\times10^6\,\mathrm{K}$.The typical temperature at this phase is 10^6–$10^7\,\mathrm{K}$, and the SNR strongly emits X-rays, mainoy through thermal bremsstrahlung. When $T < 10^6\,\mathrm{K}$, some ions start to recombine to form atoms, which show emission lines and lead to strong cooling of the gas. The Sedov-Taylor phase is terminated when radiative losses behind the outward moving blast wave start to become important and a thin shell of cooled gas forms near this outer boundary.

1.3. *The Radiative Phase (or Snowplow Phase)*

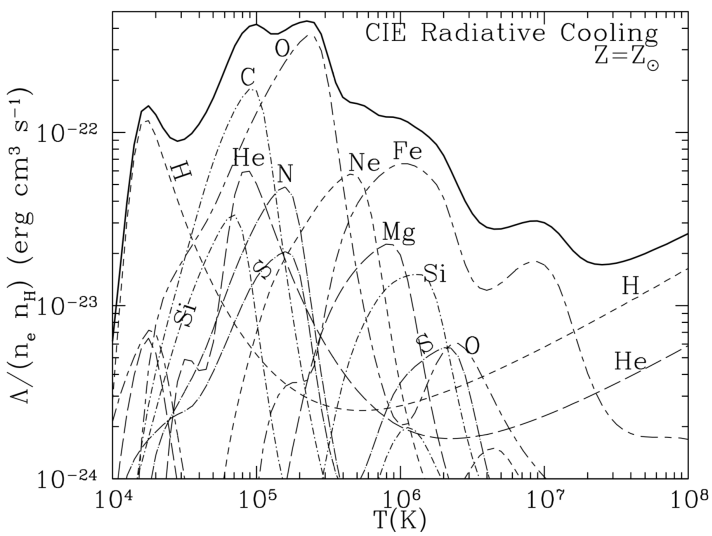

Figure 2. Radiative cooling function. Fig 34.3 of Draine (2011).

The shock wave continues to cool, and once temperatures drop below about $\sim 5\times10^5\,\mathrm{K}$, electrons start recombining to form heavier elements (in particular, oxygen line emission, see Figure 2). This recombination process radiates energy much more efficiently than the thermal X-rays and synchrotron emission produced thus far, and causes the temperature immediately behind the shock frount to drop even further. Pressured by the hot gas, the cooled gas will be compressed, forming a high-density shell. With efficient radiative cooling, the expansion can no longer be considered adiabatic and the euqation of motion will eventually be governed by the conservation of momentum.

Assuming that the momentum-conserving phase begins at $t = t_0$ and the momentum of the shell, $\mathcal{P}_0 = M_s v$, remains constant hereafter with the value of

$$\mathcal{P}_0 = \left(\frac{4\pi}{3}R_0^3\rho_0\right)\dot{R}_0$$

We have the equation of motion after t_0:

$$\mathcal{P}_0 = \left(\frac{4\pi}{3}R^3\rho_0\right)\dot{R} \tag{10}$$

This can be integrated with respect to time to give

$$\frac{1}{4}R^4 = \frac{3\mathcal{P}_0}{4\pi\rho_0}t + C$$

where

$$C = \frac{1}{4}R_0^4 - R_0^3\dot{R}_0 t_0$$

The solutions to the equation of motion are

$$R = R_0\left[1 + \frac{4\dot{R}_0(t-t_0)}{R_0}\right]^{1/4} \tag{11}$$

$$\dot{R} = \dot{R}_0\left[1 + \frac{4\dot{R}_0(t-t_0)}{R_0}\right]^{-3/4} \tag{12}$$

For $t \gg t_0 + R_0/\dot{R}_0$:

$$R \propto t^{1/4}$$
$$\dot{R} \propto t^{-3/4}$$

The shell propagates with constant radial momentum and piles up the ambient material like a *snowplow*. Towards the end of this phase, the expansion velocity becomes sonic or subsonic with respect to the interstellar sonic or magnetosonic speed, and the remaining energy of the ancient supernova remnant is finally dissipated through turbulent cascade.

1.4. *Merger with the ISM*

Occurs when $v_s = c_s$ is the sound speed.

2. STELLAR WIND

A massive O-star will spend ∼90% of its life on the main sequence, during which it blows a fast, tenuous wind with velocity $v_w = 1500$–$3000\,\mathrm{km\,s^{-1}}$, with $\dot{M} = 5\times10^{-7}$–$10^{-5}\,M_\odot\,\mathrm{yr}$. In the subsequent red supergiant (RSG) phase the mass loss rate increases to as much as $\dot{M} = 10^{-4}\,M_\odot\,\mathrm{yr}$. Stars more massive than $60M_\odot$ can even have extremely high mass loss rates, often accompanied by violent bursts of mass loss. These are so-called Luminous Blue Variables (LBVs).

Most massive stars will explode in the RSG phase, but the highest mass stars may enter the Wolf-Rayet star phase. In this phase the stars have lost all of their hydrogen envelopes. Their mass loss rates are high, $\dot{M} \approx 10^{-5}$–$10^{-4}\,M_\odot\,\mathrm{yr}$, with a high wind velocity $v_w = 1000$–$2000\,\mathrm{km\,s^{-1}}$. This results in a tenuous wind, which will sweep up the RSG wind into a dense shell.

Progenitors that were part of a binary system may even have had more complicated mass loss histories. In some cases a common envelope phase may have existed, which could also have lead to a stellar merger. In these case the circumstellar medium structure may be rather complex. See for example the binary evolution model that can explain some of the features of the circumstellar medium of SN 1987A.

2.1. *Interstellar Bubbles*

All hot stars, be they either young and massive or old low-mass objects at the cores of planetary nebulae, produce fast winds driven by **radiation pressure**. Radiation from the central star is scattered by the atmosphere, so that the momentum carried in the radiation field,L_*/c, is transferred to the atmospheric gas, to produce an outflowing wind. If v_w is the terminal velocity attained by the wind, and $\dot{M}_w$ its mass flux:

$$\dot{M}_w v_w = \frac{\eta L_*}{c} \tag{13}$$

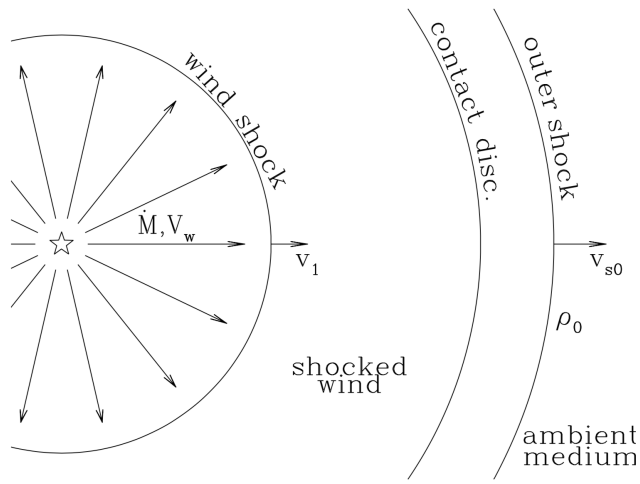

Figure 3. Structure of the bubble produced by a fast stellar wind (not to scale). Fig 38.1 of Draine (2011).

Here, the factor η accounts for the fact that each photon may be scattered many times before it escapes, enhancing the total amount of momentum that can be deposited. This process is limited, since the wind cannot carry away more energy than is produced by the star:

$$\frac{1}{2}\dot{M}_w v_w^2 < L_* \tag{14}$$

In stars, the full application of the radiatively driven wind theory shows that $\eta \sim 3$–4. In such radiatively driven winds, the outflow velocity is a factor $1 \lesssim \epsilon \lesssim 3$ times the escape velocity at the base of the outflow

$$v_w = \epsilon \left(\frac{GM_*}{r_*}\right)^{1/2} \sim 437\epsilon \left(\frac{M_*/M_\odot}{R_*/R_\odot}\right)^{1/2} \mathrm{km\,s^{-1}} \tag{15}$$

Because $v_w \gg c_s$, the wind will initially drive a shock into the H II region – the outer shock in Figure 3 – with radius R_s and velocity v_s. The shocked ambient medium accumulates in a shell behind the outer shock front.

When the dynamical pressure of the wind drops to the thermal pressure of the surrounding gas, a second shock front will form to slow the stellar wind down (analogous to the reverse shock in the early supernova evolution). A contact discontinuity separates the shocked ambient medium from the shocked wind material. If there were no instabilities, and diffusion can be neglected, these two regions would not mix. The innermost region contains the unshocked stellar wind.

At very early times, the wind undergoes essentially free expansion, with $R_s \approx v_w t$ until the wind mass and swept-up mass are comparable: $\dot{M}t \approx (4\pi/3)\rho_0(v_w t)^3$. The free expansion ends at a time

$$t_0 \approx \left(\frac{3\dot{M}}{4\pi\rho_0 v_w^3}\right)^{1/2} = 2.54 n_3^{-1/2} \dot{M}_{-6}^{1/2} v_{w,8}^{-3/2}\,\mathrm{yr} \tag{16}$$

where $v_w \equiv 10^3 v_{w,8}\,\mathrm{km\,s^{-1}}$. This is such a short time that the free expansion phase is unlikely to be observed.

After that,

- The **shocked wind material** is very hot, and the density is low. As a result, radiative losses from the shocked wind region will be negligible — this is the energy-conserving phase. There will be a long phase during which the shocked wind material loses energy primarily by continuing to do work on the dense shell.

- In contrast, the swept-up interstellar medium gas is denser, and over most of the evolution of the bubble, is not shocked to very high temperature. The **outer shock** is radiative and isothermal at the temperature ($\sim 10^4$ K) of the preshock gas, which is kept ionized by the photons from the central star. As the zone with shocked ISM cools and collapses into a thin shell, the expansion briefly slows, but then resumes as the pressure from the hot gas in the interior continues to act.

When v_s falls to the ambient sound speed, $\sim 15\,\mathrm{km\,s^{-1}}$ in an H II region, the outer shock will disappear ($t \sim 3 \times 10^4\,\mathrm{yr}$), with the pressure pulse propagating away as an acoustic wave. Unlike the outer shell, the hot gas in the interior does not make a sudden transition from predominantly hot to predominantly cold. The wind material near the contact discontinuity cools rapidly to $T \approx 10^4\,\mathrm{K}$, resulting in a thin layer of cooled stellar wind material in contact with the cool shell of ISM material, even while much or most of the shocked stellar wind material is still hot.

2.2. *Interacting Stellar Winds*

In constrast to the interstellar bubble case where a stellar wind runs into a stationary, constant density medium, the interacting winds process considers the possibility that both media have a density and velocity structure. This mechanism was first proposed to explain the origin of planetary nebulae and has since been applied to a variety of other astrophysical phenomena (e.g., ring nebulae around massive stars, supernovae, young star outflows, relativistic jets in AGN).

We consider collision of two flows:

- AGB phase wind ($v_w \sim 10\,\mathrm{km\,s^{-1}}$, $\dot{M} \sim 10^{-4}\,M_\odot\,\mathrm{yr^{-1}}$).
- Central Star of planetary nebula phase wind ($v_w \sim 2000\,\mathrm{km\,s^{-1}}$, $\dot{M} \sim 10^{-7}\,M_\odot\,\mathrm{yr^{-1}}$).

The interaction of the fast winder with the pre-existing slower sind produces a shell containing both AGB ejecta and shocked fast wind material. At the same time that this shell accumulation is occuring, the ultraviolet radiation from the increasingly host central star will photoionize the shell. The balance between the photoionization and recombination rates in the material leads to the prominent Hα emission of planetary nebulae. The density also tend to be low enough that the forbideen line radiation such as the [O III] λ5007 line is prominent.

REFERENCES

Dopita, M. A., & Sutherland, R. S. 2003, Astrophysics of the diffuse universe

Draine, B. T. 2011, Physics of the Interstellar and Intergalactic Medium

Kwok, S. 2007, Physics and Chemistry of the Interstellar Medium

Rosswog, S., & Brüggen, M. 2011, Introduction to High-Energy Astrophysics

Tielens, A. G. G. M. 2005, The Physics and Chemistry of the Interstellar Medium

Vink, J. 2012, A&A Rv, 20, 49